DISCRETE MATHEMATICS PRESENTATION

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Question No. 01

Define propositional Logic with example

Ans:

A propositional is a declarative sentence that is a sentence that declares a fact that is either true or false but not both. A proposition consist of propositional variables and connectives. We denote the propositional variables by capital letters (A,B,etc). The connective connect the propositional variables . Some example of propositional are given below.

1. **1+1=3(False):**

This is a declarative sentence and it clearly states something false. Since a proposition must have a define truth value (either true or false), this is a valid proposition and the truth value is false.

2. **1+1= 2(True):**

This is another declarative sentence that is clearly true. It has a definite truth value, which is true. Therefore, it is a valid proposition

Question No. 02

Define connectives uses of five kinds of connectives by trouth table

Ans:

Connectives in logic are operators that combine propositions to form more complex statements. The five common types of logical connectives are:

1. Negation (¬)
2. Conjunction (∧)
3. Disjunction (∨)
4. Implication (→)
5. Biconditional (↔)

1. Negation (¬)

P	¬P
T	F
F	T

2. Conjunction (∧)

Let P and Q be propositions. The conjunction of A and B denoted by P∧Q. The conjunction of P∧Q is true when both P and Q are true and is false otherwise. The truth table is as follows-

P	Q	P^Q
T	T	T
T	F	F
F	T	F
F	F	F

3. Disjunction (∨): Let P and Q be propositions. The disjunction of P and Q denoted by P∨Q is false when both P and Q are false and true otherwise. The truth table is as follows-

P	Q	P v Q
T	T	T
T	F	T
F	T	T
F	F	F

4. Implication (→)

Let A and B be propositions. It is denoted by P→Q. The conditional statement P→Q is false when P is true but Q is false and true otherwise. In the conditional statement P is called hypothesis and Q is called the conclusion. The truth table is as follows-

P	Q	P→Q
T	T	T
T	F	F
F	T	T
F	F	T

5. Biconditional (↔)

Let P and Q propositions. The exclusive or of P and Q is denoted by P ↔ Q. The proposition that is true when exactly one of P and Q is true and is false otherwise. The truth table is as follows-

P	Q	P↔Q
T	T	T
T	F	F
F	T	F
F	F	T

Question No. 03

Prove that P→(2^2) = (p+q)^(P) using truth table

Ans:

Hence, we can see from the truth table that, p -> (q ^ r) = (p -> q) (pr). (Proved)

p	q	r	q^r	p→(q^r)	p → q	p → r	(p → q)Λ(p → r)
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	F	F	T	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
T	T	T	T	F	F	F	F

Question No. 04

Define compound proposition and classified the compound proposition explain using trouth table.

Ans:

Compound proposition is a proposition formed by combining two or more simple propositions. There are three types of compound proposition-

1) Tautology 2) Contradiction 3) Contingency

1. Tautology:

A compound proposition that is always true regardless of the truth values of its components.Example: (Law of the Excluded Middle).

p	p¬	p v p¬
T	F	T
F	T	T

2. Contradiction

A compound proposition that is always false regardless of the truth values of its components.Example: (Law of Contradiction).

p	¬p	p ^ ¬p
T	F	F
F	T	T

3.Contingency

A compound proposition that is sometimes true and sometimes false depending on the truth values of its components.

P	Q	P→ Q
T	T	T
T	F	F
F	T	T
F	F	T

Question Number 05.

What kind of compound proposition of this $(\sim p \wedge (p \rightarrow q)) \rightarrow \sim p$?

Ans:

Hence, every value of propositional variables is "TRUE" in the truth table, so it is a tautology.

p	q	¬p	¬q	p→ q	¬p ∧(p→q)	(¬p∧ (p→ q) → ¬q)
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Question 06:

Prove De Morgan Law in Discrete by using the truth table.

Ans:

De Morgan's laws state that:

1. ¬(P∧Q)=(¬P∨¬Q) [First Law]
2. ¬(P∨Q)=(¬P∧¬Q) [second Law]

We will prove these laws using truth tables:

1. Proof of First Law:

P	Q	P^Q	¬(P^Q)	¬P	¬Q	¬P V ¬Q
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	F	F	T
F	F	F	T	F	T	T

2.Proof of Second Law:

P	Q	P V Q	¬(P V Q)	¬P	¬Q	¬P V ¬Q
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Question 07:

Show that $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ logical equivalence.

Ans:

Hence, the truth tables of each statement have same truth values, so, $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ logical equivalence.

Question 07:

Ans: Truth tables values

p	q	r	P → r	q → r	(p → r)v(q → r)	p Λq	(p Λq) →r
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	F	T
T	F	F	F	T	T	F	T
F	T	T	T	T	T	F	T
F	T	F	T	F	T	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

Question 08:

Define relation. Give five examples of reflexive relations.

Ans:

A relation in mathematics is a set of ordered pairs that defines a relationship between elements of two sets. If a relation exists from a set A to itself, it is called a relation on set A and a subset of A and a subset of A × A.

Five Examples of Reflexive Relations:

1.Equality relation (=) set : A={1,2,3} .relation : R ={(1,1),(2,2),(3,3)} Here every element is related to itself.

2.Less than or equal to (≤) on real numbers: .set : A=R(set of real numbers) .Relation: R={(x , y) | x ≤ y } Since x ≤ x is always true for all x € R , the relation is reflexive.

3. Divisibility Relation: Set: A = {1, 2, 3, 4} Relation: R = {(a, b) | a divides b} Since every number divides itself, the relation is reflexive.

4. Subset Relation (<=) on the Power set: set A = {(a), (2), (4, 23) Relation: R = {(x, y) | x <= y} Since every set is a subset of itself, this relation is reflexive.

5. Universal Relation: set A = {1, 2, 33 Relation: R = A×A Since it is the universal relation, it is reflexive.

Question 09:

Let A be a set of students and R be a binary relation of A such that (a,b) is in R if and only if a lives in the same hall as b. Is R reflexive? Justify your answer.

Ans:

A relation R on a set A is reflexive if aRa for every a∈A, that is, if (a,a)∈R for every a∈A we can call it reflexive.

In our case R is define by the property that for any students a and b in A, (a , b) € R if and only if a and b live in the same hall. notice that for any student a , the statement ” a live in the same hall as a “ is true because every student is always in the same hall as themselves. Therefor, we have , a € A , (a , a) € R .

Thus, R is reflexive.

Question 10:

How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?

Ans:

Here, n = 10 (total number of players)

r = 5 (the number of players to select)

We know the combination formula:

C(n, r) = {n!}/{r!(n-r)!}

C(10,5) = {10!}/{5!5!} =

{10 × 9 × 8 × 7 × 6 × 5!}/{5! × 5 × 4 × 3 × 2 × 1}

= {30× 240}/{120}

= 252

Therefore, there are 252 ways.

Question 11.

A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Ans:

Here, n = 12 (the total number of offices)

r = 2 (the number of offices to be assigned)

Using the permutation formula:

$P(n, r) = \{n!\}/\{(n-r)!\}$

$P(12,2) = \{12!\}/\{(12-2)!\}$

$= \{12!\}/\{10!\}$

$= \{12 \times 11 \times 10!\}/\{10!\}$

$= 132$

Thus, there are 132 ways.